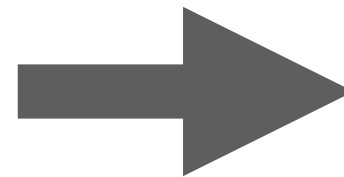
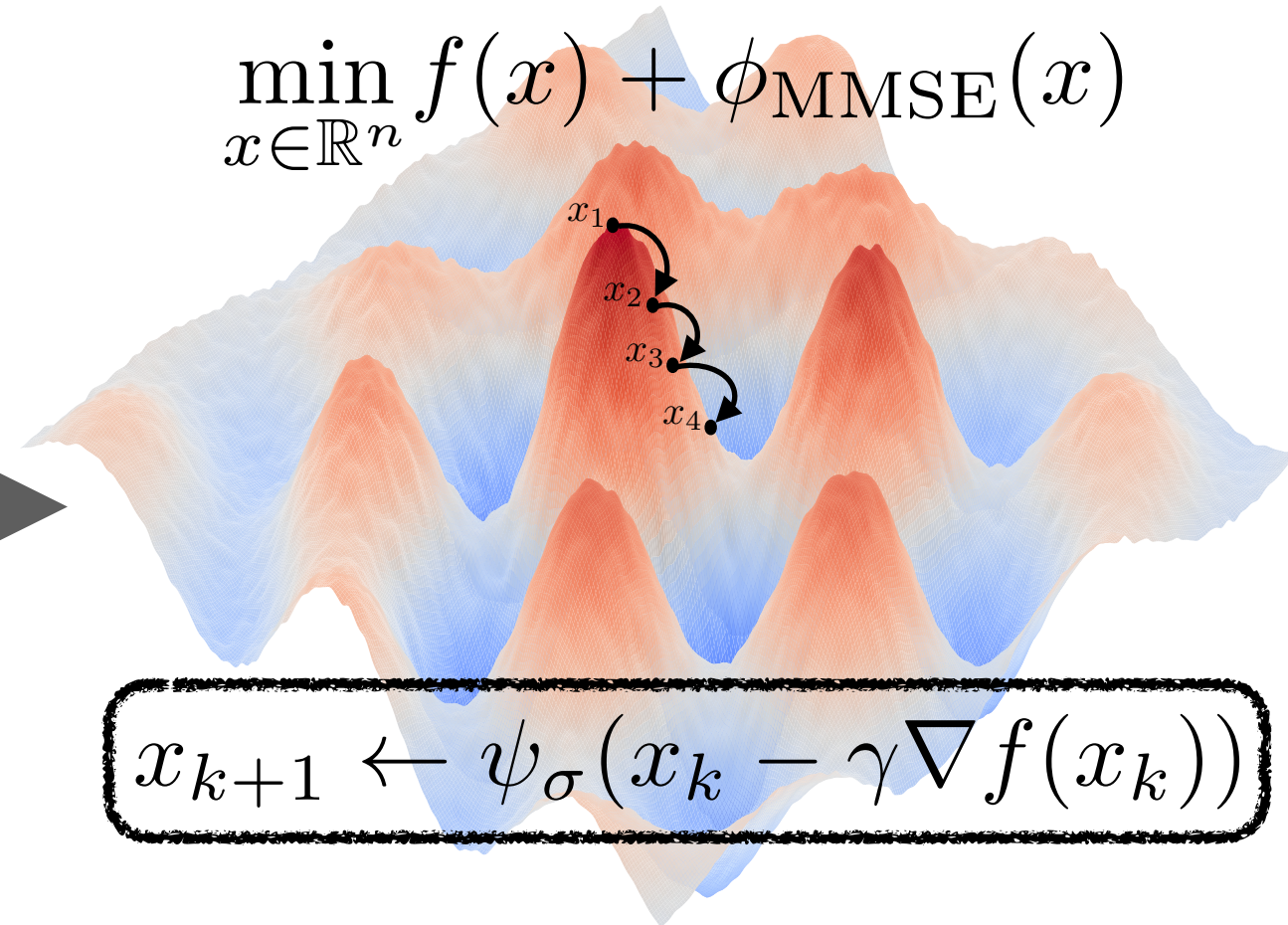
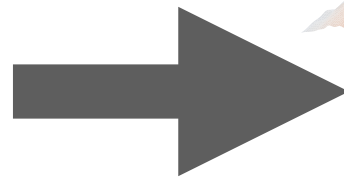
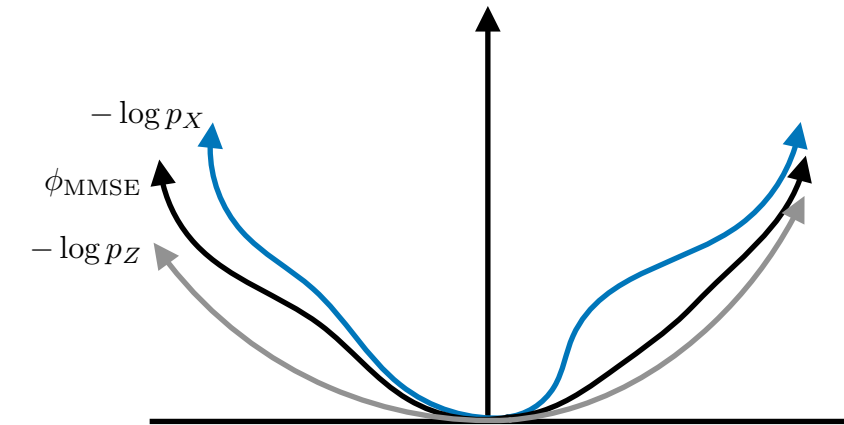


$$y := Ax + \epsilon, \quad z := x + \epsilon$$



$$\psi_{\sigma}(z) := \mathbb{E}[x \mid z] = \text{prox}_{\phi_{\text{MMSE}}}(z)$$



$\therefore$ 1-weak convexity of ϕ_{MMSE}
 $\implies \mathcal{O}(1/\sqrt{K})$ Convergence.